PRACTICAL NO 8 : IMAGE MATTING AND COMPOSTING

```
import cv2
import numpy as np
def estimate_alpha(image,trimap):
    #placeholder function replace with your matting algorithm implementation
    #this example simply sets alpha  values based on
trimap(eg.foreground=1,background=0,unknown=interpolated)
    alpha=np.zeros_like(trimap,dtype=np.float32)
    alpha[trimap==255]=1.0 #foreground
    alpha[trimap==0]=0.0 #background
    alpha[(trimap>0)&(trimap<255)]=0.5 #interploted
    return alpha

def image_matting(image,trimap):
    #convert image and trimap to float32
    image=image.astype(np.float32)/255.0
    trimap=trimap.astype(np.float32)/255.0

    #estimate alpha matte using a matting algorithm
    #replace this with your designed matting algorithm
    alpha=estimate_alpha(image,trimap)

    #clip alpha values to [0,1]
    alpha =np.clip(alpha,0,1)

    return alpha

 # composit foreground and background using alpha matte
def composit_foreground_background(foreground,background,alpha):
    background=cv2.resize(background(foreground.shape[1],foreground.shape[0]))
    alpha=np.stack((alpha,alpha,alpha),axis=2)
    composited_image=alpha*foreground+(1-alpha)*background
```

```
    return composited_image


#example
if __name__=="__main__":
    foreground=cv2.imread("D:\TYDS07\model.jpg")
    background=cv2.imread("D:\TYDS07\model.jpg")
    trimap=cv2.imread("D:\TYDS07\model.jpg",cv2.IMREAD_GRAYSCALE)

    #perform image matting
    alpha=image_matting(foreground,trimap)

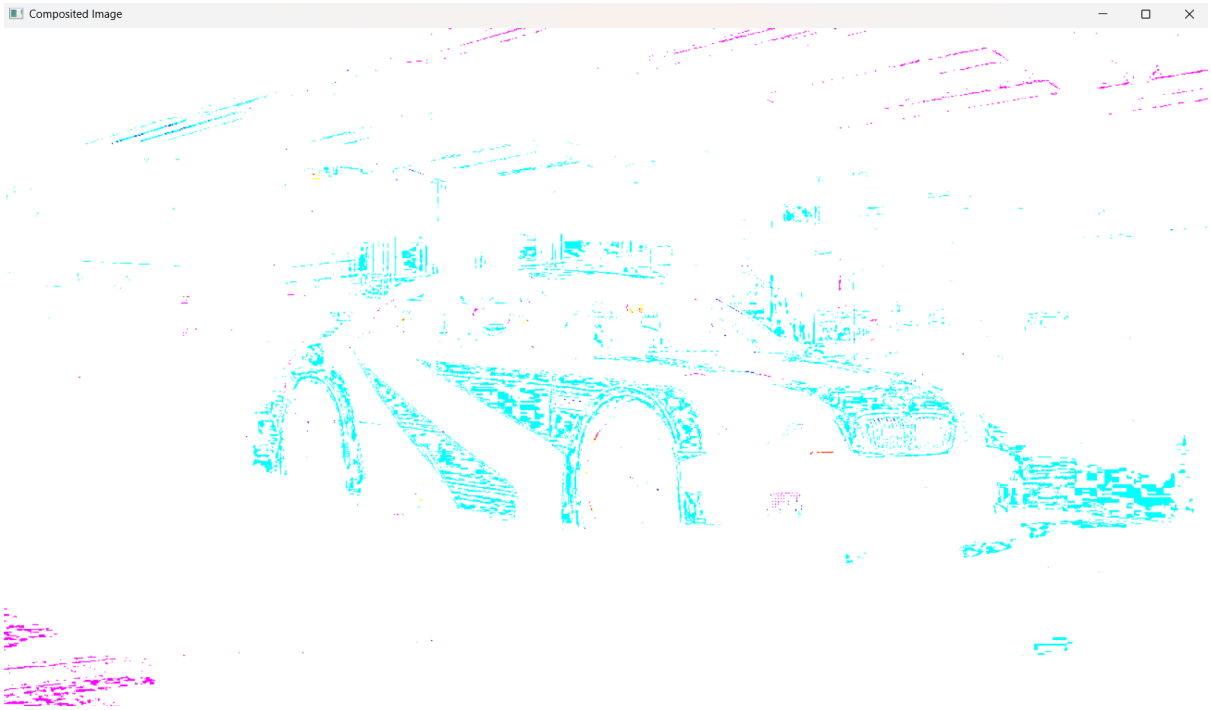
    #perform comosting
    composited_image=composit_foreground_background(foreground,background,alpha)

    #display result
    cv2.imshow('Composited Image',composited_image)
    cv2.waitKey()
    cv2.destroyAllWindows()
```

Output :




Alpha Matte


Composited Image